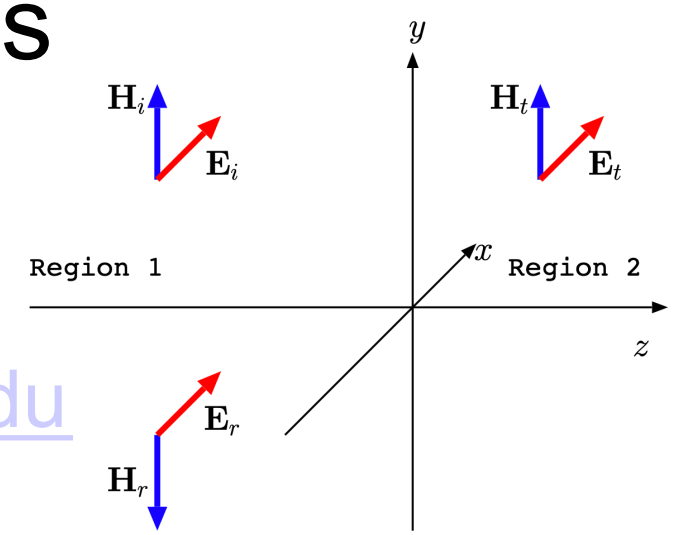
Lectures 25-26

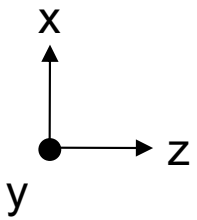
Reflection and Transmission
Standing Waves

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Normal incidence plane wave



Medium 1

$\sigma_1, \varepsilon_1, \mu_1$

Medium 2

$\sigma_2, \varepsilon_2, \mu_2$

Incident (+) ⟶

$$\widetilde{E}_i(z) = \overline{E}_1^+ e^{-\bar{\gamma}_1 z}\hat{x}$$

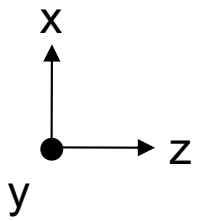
$$\widetilde{H}_i(z) = \overline{H}_1^+ e^{-\bar{\gamma}_1 z}\hat{y} = \frac{1}{\overline{\eta}_1}\overline{E}_1^+ e^{-\bar{\gamma}_1 z}\hat{y}$$

What should happen next?

z<0 | z>0

We will analyze the case J_s=0 at z=0 (no applied surface current).

Reflected and Transmitted Waves are Generated

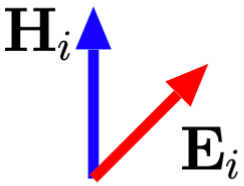


Medium 1

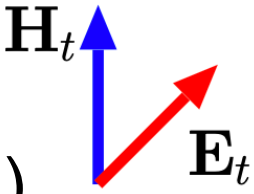
$\sigma_1, \varepsilon_1, \mu_1$

Medium 2

$\sigma_2, \varepsilon_2, \mu_2$



Incident (+z) ⟶ ⟶ Transmitted (+z)

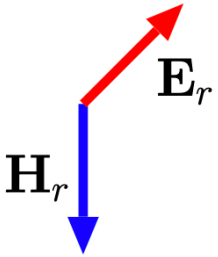


$$\widetilde{E}_i(z) = \overline{E}_1^+ e^{-\bar{\gamma}_1 z}\hat{x},\ \widetilde{H}_i(z) = \frac{1}{\overline{\eta}_1}\overline{E}_1^+ e^{-\bar{\gamma}_1 z}\hat{y}$$

$$\widetilde{E}_t(z) = \overline{E}_2^+ e^{-\bar{\gamma}_2 z}\hat{x},\ \widetilde{H}_t(z) = \frac{1}{\overline{\eta}_2}\overline{E}_2^+ e^{-\bar{\gamma}_2 z}\hat{y}$$

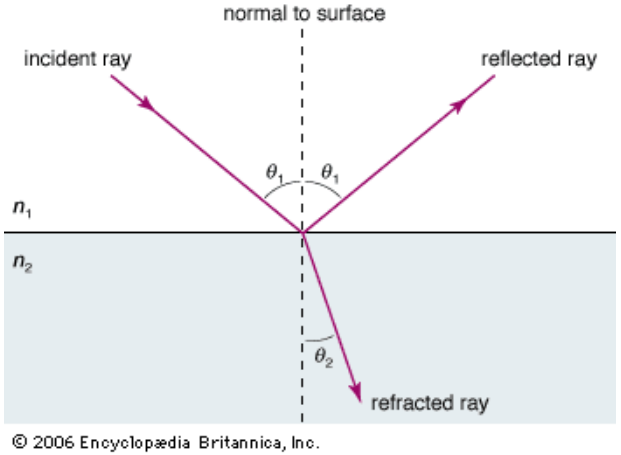
Reflected (-z) ⟵

$$\widetilde{E}_r(z) = \overline{E}_1^- e^{+\bar{\gamma}_1 z}\hat{x},\ \widetilde{H}_r(z) = \frac{-1}{\overline{\eta}_1}\overline{E}_1^- e^{+\bar{\gamma}_1 z}\hat{y}$$

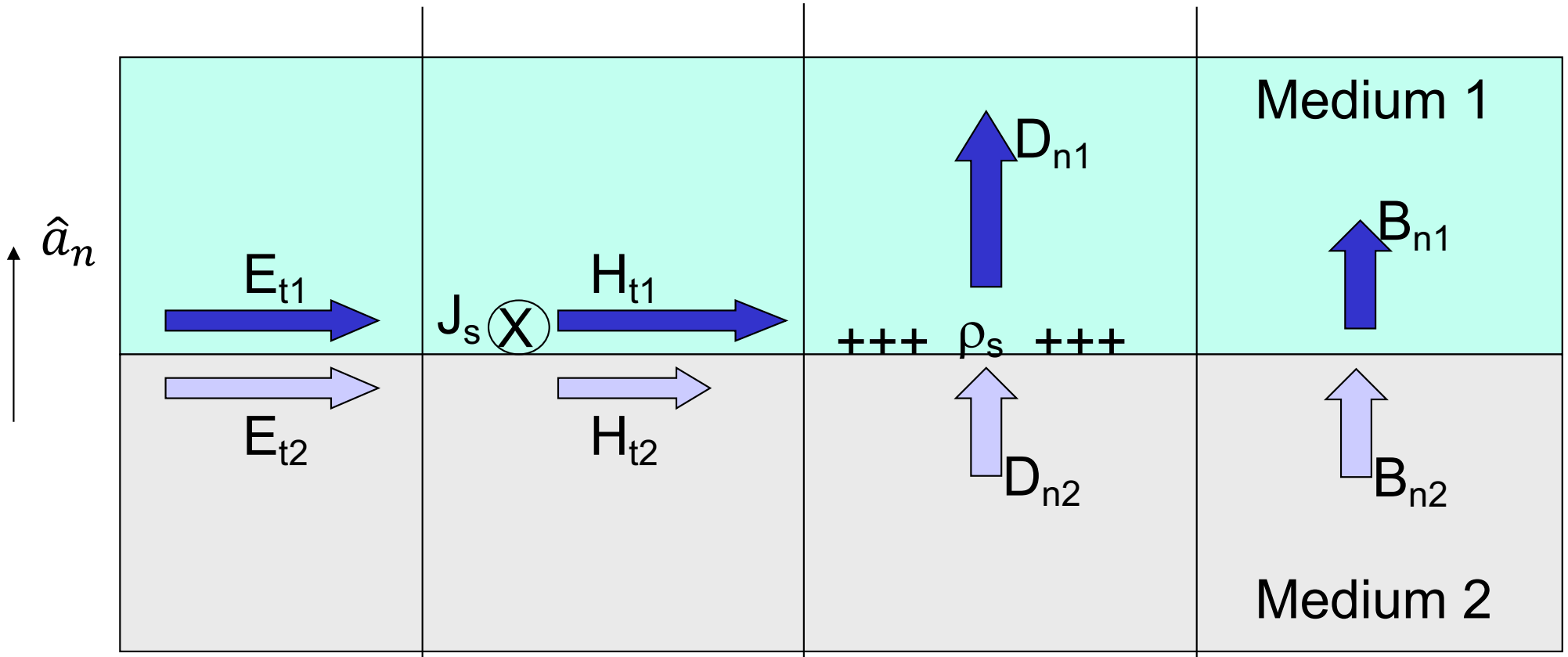


z<0 z>0

At z=0, $\mathbf{J}_S$=0



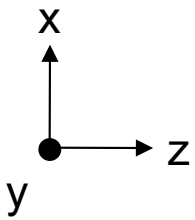
Apply Boundary Conditions



$E_{t1} - E_{t2} = 0$ | $|H_{t1} - H_{t2}| = |\vec{J}_s|$ | $D_{n1} - D_{n2} = \rho_s$ | $B_{n1} - B_{n2} = 0$

$\hat{a}_n \times (\vec{E}_1 - \vec{E}_2) = 0$ | $\hat{a}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s$ | $\hat{a}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s$ | $\hat{a}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0$

Apply Boundary Conditions



Medium 1

$\sigma_1, \varepsilon_1, \mu_1$

Incident (+z) →

Reflected (-z) ←

$$\tilde{E}_{1x\,total}(z) = \overline{E}_1^{+} e^{-\bar{\gamma}_1 z} + \overline{E}_1^{-} e^{+\bar{\gamma}_1 z}$$

$$\tilde{H}_{1y\,total}(z) = \frac{1}{\overline{\eta}_1}\left(\overline{E}_1^{+} e^{-\bar{\gamma}_1 z} - \overline{E}_1^{-} e^{+\bar{\gamma}_1 z}\right)$$

Medium 2

$\sigma_2, \varepsilon_2, \mu_2$

Transmitted (+z) →

$$\tilde{E}_{2x\,total}(z) = \overline{E}_2^{+} e^{-\bar{\gamma}_2 z}$$

$$\tilde{H}_{2y\,total}(z) = \frac{1}{\overline{\eta}_2}\overline{E}_2^{+} e^{-\bar{\gamma}_2 z}$$

B.C. $E_{t1}=E_{t2}$ $H_{t1}-H_{t2}=J_s=0$

$$\tilde{E}_{1x\,total}(0^-) = \overline{E}_1^{+} + \overline{E}_1^{-} = \overline{E}_2^{+} = \tilde{E}_{2x\,total}(0^+)$$

$$\tilde{H}_{1y\,total}(0^-) = \frac{1}{\overline{\eta}_1}\left(\overline{E}_1^{+} - \overline{E}_1^{-}\right) = \frac{1}{\overline{\eta}_2}\overline{E}_2^{+} = \tilde{H}_{2y\,total}(0^+)$$

Solve the Equations

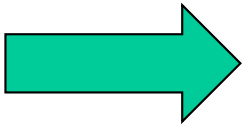
Define

Reflection Coefficient: $\Gamma = \dfrac{\bar{E}_1^-}{\bar{E}_1^+}$

Transmission Coefficient: $\tau = \dfrac{\bar{E}_2^+}{\bar{E}_1^+}$

$$\overline{E}_1^+ + \overline{E}_1^- = \overline{E}_2^+$$

$$\frac{\overline{\eta}_2}{\overline{\eta}_1}\left(\overline{E}_1^+ - \overline{E}_1^-\right) = \overline{E}_2^+$$

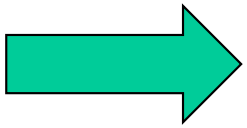


$$\bar{E}_1^+(1+\Gamma) = \tau\bar{E}_1^+$$

$$\frac{\bar{\eta}_2}{\bar{\eta}_1}\bar{E}_1^+(1-\Gamma) = \tau\bar{E}_1^+$$

Solve the Equations

$$1 + \Gamma = \tau$$
$$\frac{\bar{\eta}_2}{\bar{\eta}_1}(1 - \Gamma) = \tau$$



$$1 + \Gamma = \frac{\bar{\eta}_2}{\bar{\eta}_1}(1 - \Gamma)$$

Reflection Coefficient:

$$\therefore \Gamma = \frac{\bar{E}_1^-}{\bar{E}_1^+} = \frac{\bar{\eta}_2 - \bar{\eta}_1}{\bar{\eta}_1 + \bar{\eta}_2}$$

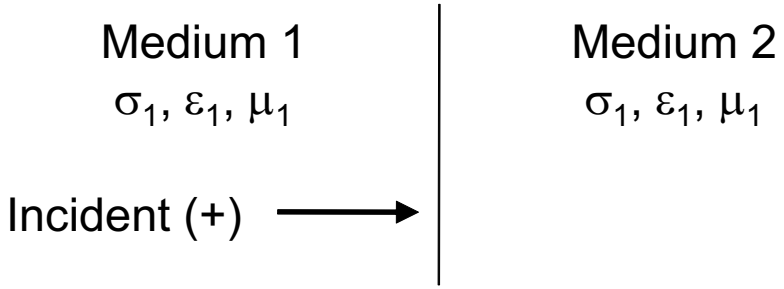
$$\tau = 1 + \Gamma$$

Transmission Coefficient:

$$\therefore \tau = \frac{\bar{E}_2^+}{\bar{E}_1^+} = 1 + \Gamma = \frac{2\bar{\eta}_2}{\bar{\eta}_1 + \bar{\eta}_2}$$

Challenge Question: Special cases

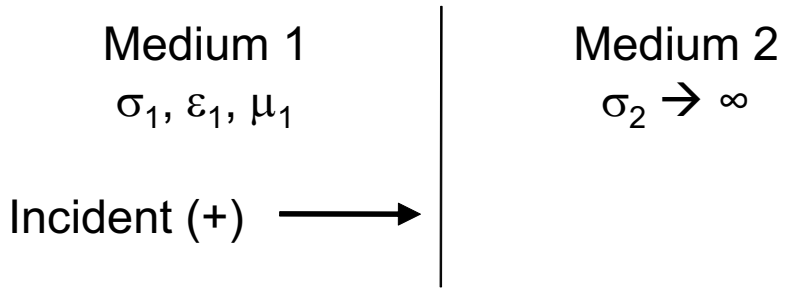
- If $\eta_2=\eta_1$, we expect:



(a) $\Gamma=0$, $\tau=1$ (0% reflected, 100% transmitted)
(b) $\Gamma=0$, $\tau=-1$ (0% reflected, 100% transmitted)
(c) $\Gamma=1$, $\tau=0$ (100% reflected, 0% transmitted)
(d) $\Gamma=-1$, $\tau=0$ (100% reflected, 0% transmitted)
(e) cannot be determined

Challenge Question: Special cases

- If medium 2 is a perfect conductor, we expect:



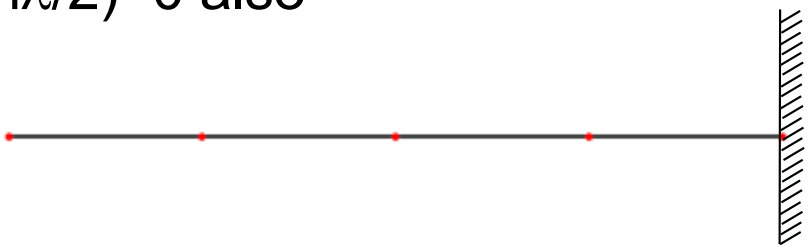
(a) Γ=0, τ=1 (0% reflected, 100% transmitted)
(b) Γ=0, τ=-1 (0% reflected, 100% transmitted)
(c) Γ=1, τ=0 (100% reflected, 0% transmitted)
(d) Γ=-1, τ=0 (100% reflected, 0% transmitted)
(e) cannot be determined

Practical application: _____

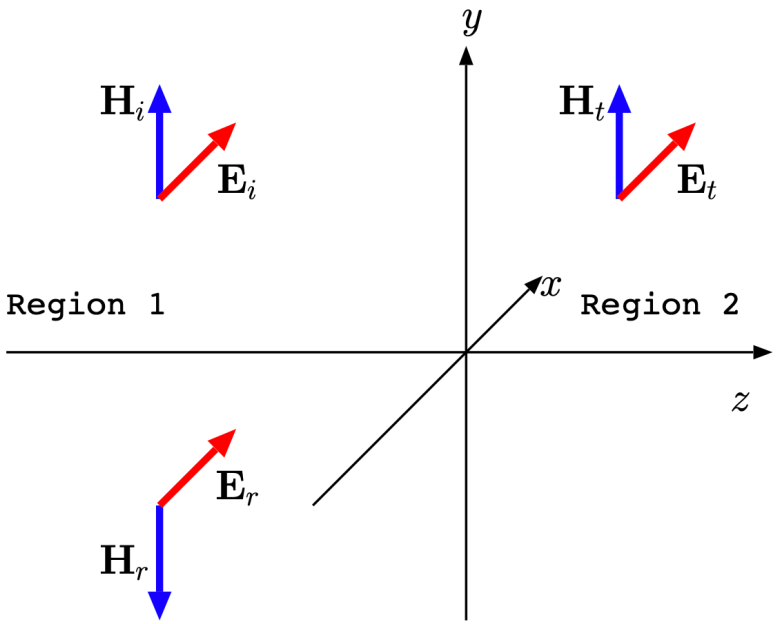
Special Cases of Reflection/Transmission

$$\Gamma \equiv \frac{\overline{E}_1^-}{\overline{E}_1^+} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}, \tau \equiv \frac{\overline{E}_2^+}{\overline{E}_1^+} = 1 + \Gamma$$

- Impedance matched: $\eta_2=\eta_1$, then $\Gamma=0$, $\tau=1$
  - No reflection, entire wave is transmitted
- Perfect dielectrics: $\sigma_1=\sigma_2=0$, then Γ and τ are real since η_1 and η_2 are real
- Perfect conductor for medium 2: $\eta_2 \rightarrow 0$, then $\Gamma=-1$, $\tau=0$
  - No transmission, entire wave is reflected. If medium 1 is a perfect dielectric, a standing wave is set up there since reflected wave perfectly cancels input wave at many nodes: E(z=0)=0 always, but if $\sigma_1=0$, then E(z=-mλ/2)=0 also



Example: Normal incidence of a uniform plane wave onto a material medium



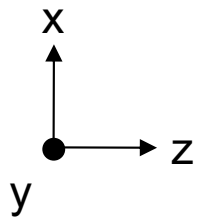
free space $\epsilon_r = \frac{9}{4}, \mu_r = 1$

For a uniform plane wave having the electric field $\widetilde{\boldsymbol{E}}_i = \sqrt{120\pi} e^{-j\beta z} \hat{a}_x$ V/m incident on the interface z = 0 from region 1. Determine the average Poynting vectors $\langle \boldsymbol{S}_i \rangle$, $\langle \boldsymbol{S}_r \rangle$, and $\langle \boldsymbol{S}_t \rangle$ of the incident, reflected, and transmitted fields.

Standing Wave

- https://courses.grainger.illinois.edu/ece329/secure/notes/reflectedwaves.html

Standing Waves



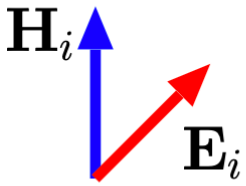
- The reflection from a PEC gives $\Gamma = -1$

Medium 1

$\sigma_1, \varepsilon_1, \mu_1$

Medium 2

$\sigma_2 \to \infty$



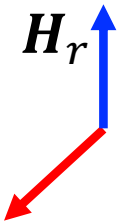
Incident (+z) ⟶

$$\widetilde{E}_i(z) = \overline{E}_1^+ e^{-\bar{\gamma}_1 z}\hat{x},\ \widetilde{H}_i(z) = \frac{1}{\overline{\eta}_1}\overline{E}_1^+ e^{-\bar{\gamma}_1 z}\hat{y}$$

Reflected (-z) ⟵

$$\widetilde{E}_r(z) = \overline{E}_1^- e^{+\bar{\gamma}_1 z}\hat{x},\ \widetilde{H}_r(z) = \frac{-1}{\overline{\eta}_1}\overline{E}_1^- e^{+\bar{\gamma}_1 z}\hat{y}$$

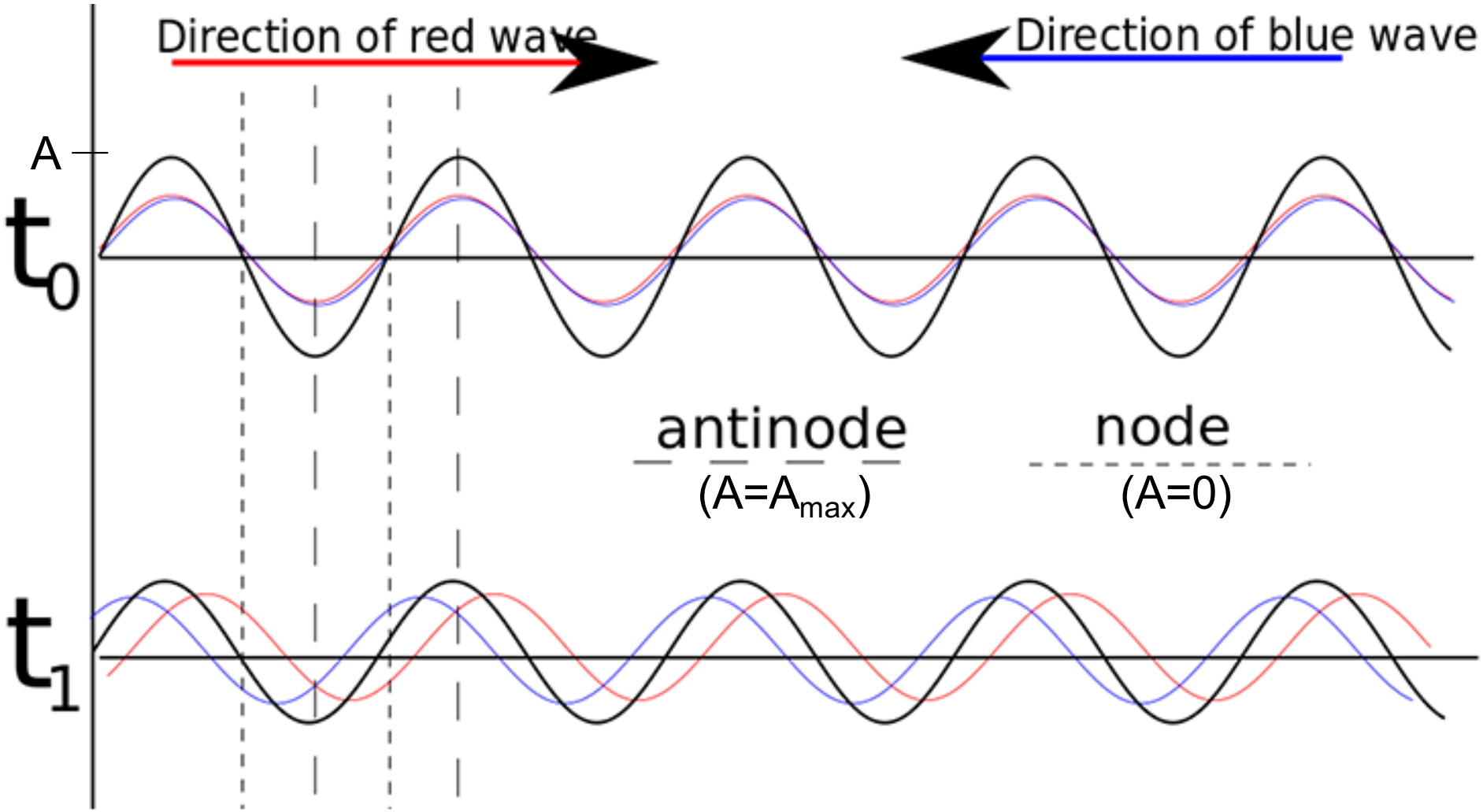
$$= -\bar{E}_1^+ e^{+\bar{\gamma}_1 z}\hat{x} \qquad = \frac{1}{\bar{\eta}_1}\bar{E}_1^+ e^{+\bar{\gamma}_1 z}\hat{y}$$



z<0 z>0

At z=0, $\mathbf{J}_S$=0

Standing Waves



Standing Waves

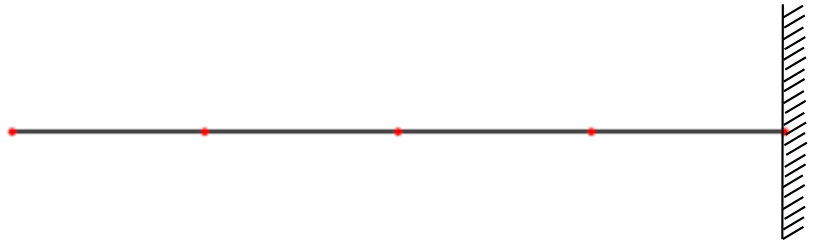
- The reflection from a PEC gives $\Gamma = -1$
- If region 1 is a perfect dielectric, we get standing waves whereby the input and reflected wave add destructively at specific points in space (vs. travelling waves $f(t \pm z/v)$)

$$\widetilde{E}_{1x\ total}(z) = \overline{E}_1^+ e^{-\bar{\gamma}_1 z} + \overline{E}_1^- e^{+\bar{\gamma}_1 z} = \overline{E}_1^+ (e^{-j\beta_1 z} - e^{+j\beta_1 z}) = \overline{E}_1^+ (-2j\sin(\beta_1 z))$$

$$\widetilde{H}_{1y\ total}(z) = \frac{1}{\overline{\eta}_1}\left(\overline{E}_1^+ e^{-\bar{\gamma}_1 z} - \overline{E}_1^- e^{+\bar{\gamma}_1 z}\right) = \frac{\overline{E}_1^+}{\overline{\eta}_1}(e^{-j\beta_1 z} + e^{+j\beta_1 z}) = \frac{\overline{E}_1^+}{\overline{\eta}_1} 2\cos(\beta_1 z)$$

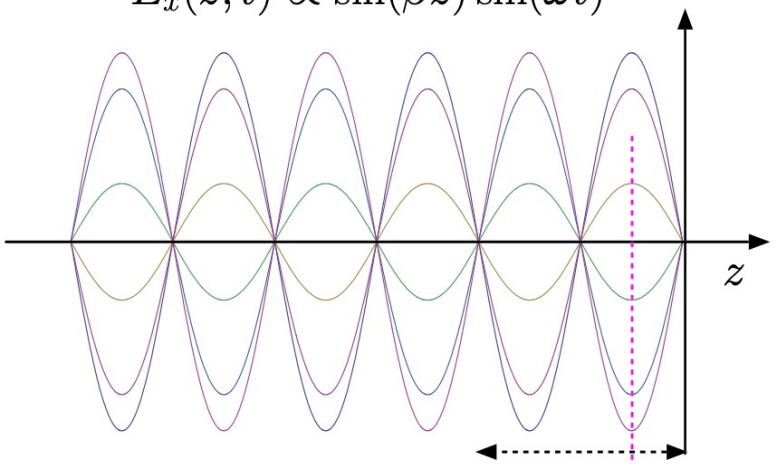
$$\vec{E}(z,t) = \hat{x}\,\mathrm{Re}[\overline{E}_1^+ (-2j\sin(\beta_1 z)) e^{j\omega t}] = \hat{x} 2\left|\overline{E}_1^+\right| \sin(\beta_1 z)\sin(\omega t + \angle \overline{E}_1^+)$$

$$\vec{H}(z,t) = \hat{y}\,\mathrm{Re}[\frac{\overline{E}_1^+}{\overline{\eta}_1} 2\cos(\beta_1 z) e^{j\omega t}] = \hat{y}\frac{2}{\eta_1}\left|\overline{E}_1^+\right| \cos(\beta_1 z)\cos(\omega t + \angle \overline{E}_1^+)$$



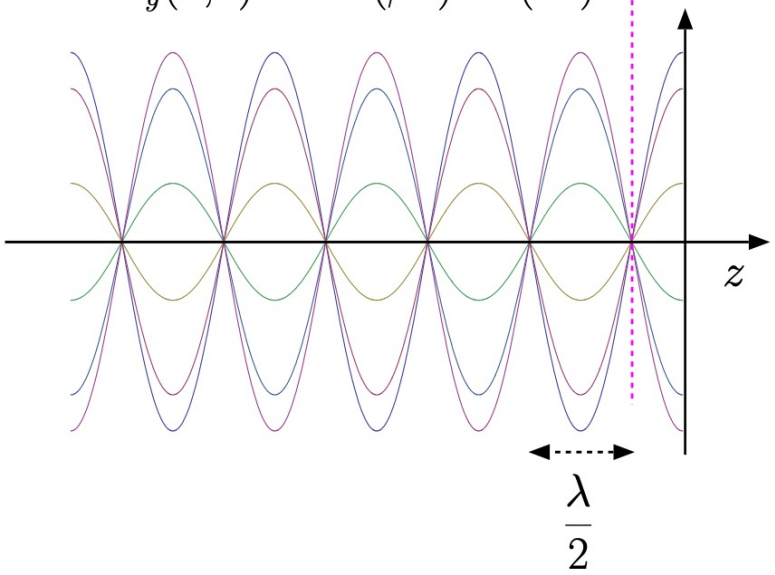
Standing Waves

$$E_x(z,t) \propto \sin(\beta z)\sin(\omega t)$$



$$\lambda = \frac{2\pi}{\beta}$$

$$H_y(z,t) \propto \cos(\beta z)\cos(\omega t)$$



$$\frac{\lambda}{2}$$

Standing Waves: on average, the net energy transport is zero

Show $\langle \boldsymbol{S} \rangle = \langle \boldsymbol{E} \times \boldsymbol{H} \rangle = 0$. Online notes (329lect26.pdf) use phasors. Here we'll use **E** and **H** explicitly (still assume σ_1=0).

$$\vec{E}(z,t) = 2\left|\overline{E}_1^+\right| \sin(\beta_1 z)\sin(\omega t + \angle \overline{E}_1^+)\hat{x}$$

$$\vec{H}(z,t) = \frac{2}{\eta_1}\left|\overline{E}_1^+\right| \cos(\beta_1 z)\cos(\omega t + \angle \overline{E}_1^+)\hat{y}$$

$$\vec{S}(z,t) = \frac{4}{\eta_1}\left|\overline{E}_1^+\right|^2 \sin(\beta_1 z)\cos(\beta_1 z)\sin(\omega t + \angle \overline{E}_1^+)\cos(\omega t + \angle \overline{E}_1^+)\hat{z}$$

$$= \frac{1}{\eta_1}\left|\overline{E}_1^+\right|^2 \sin(2\beta_1 z)\sin(2(\omega t + \angle \overline{E}_1^+))\hat{z}$$

$$\therefore \quad \left\langle \vec{S}(z,t) \right\rangle = 0$$

since $\sin(2\omega t)$ is periodic

At any moment in time, there is energy transport, but it averages to zero.

Incident wave induces surface current on PC that radiates the reflected wave

H is discontinuous at z=0, $\mathbf{H}_{t2}$=0 but $\mathbf{H}_{t1}$ oscillates:

$$\vec{H}(z,t) = \hat{y}\frac{2}{\eta_1}\left|\overline{E}_1^+\right|\cos(\omega t + \angle\overline{E}_1^+)$$

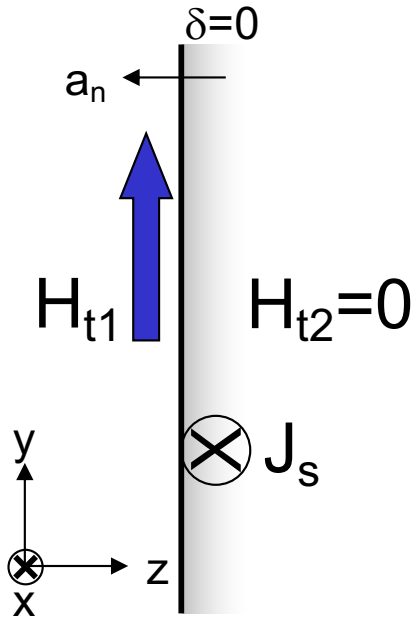
Thus, the BC's imply a surface current is induced at z=0:

$$\vec{J}_S(t) = \hat{x}\frac{2}{\eta_1}\left|\overline{E}_1^+\right|\cos(\omega t + \angle\overline{E}_1^+)$$

This surface current is now a source that can radiate a wave away from the plane in $-a_z$ (i.e. the reflected wave):

$$\vec{E}_{1reflected}(z,t) = (-\hat{x})\frac{\eta_1}{2}J_s(t-\frac{z}{v_p}) = -\hat{x}\left|\overline{E}_1^+\right|\cos(\omega t + \beta z + \angle\overline{E}_1^+)$$

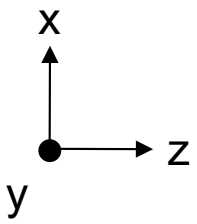
which is precisely the wave we found when we used Γ=-1. The part that radiates into $+a_z$ perfectly cancels out the incident field for z>0 which gives us E=0 in the PC as expected.



$$\hat{a}_n \times \vec{H}_{1t} = \vec{J}_S$$

Slight complication: J was created by J=σE, but E=0 at the PC. Note σ=∞.

Lectures 25-26 Summary

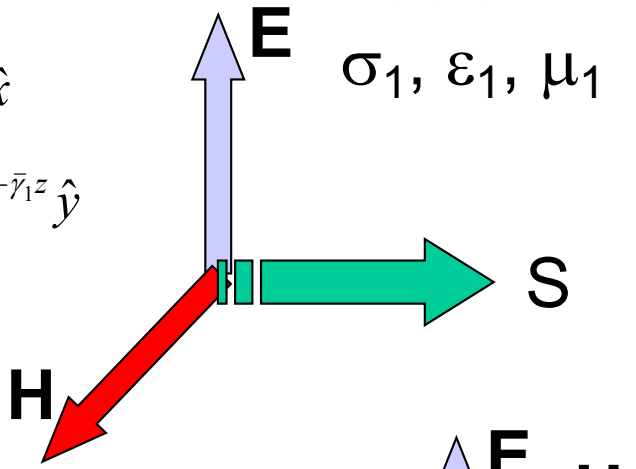


Medium 1 $\sigma_1, \varepsilon_1, \mu_1$

Medium 2 $\sigma_2, \varepsilon_2, \mu_2$

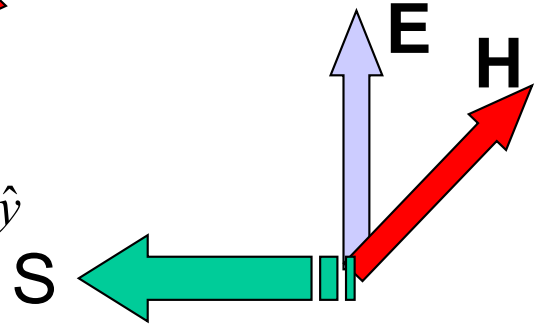
$$\tilde{E}_1(z) = \overline{E}_1^+ e^{-\bar{\gamma}_1 z}\hat{x}$$

$$\tilde{H}_1(z) = \frac{1}{\overline{\eta}_1}\overline{E}_1^+ e^{-\bar{\gamma}_1 z}\hat{y}$$



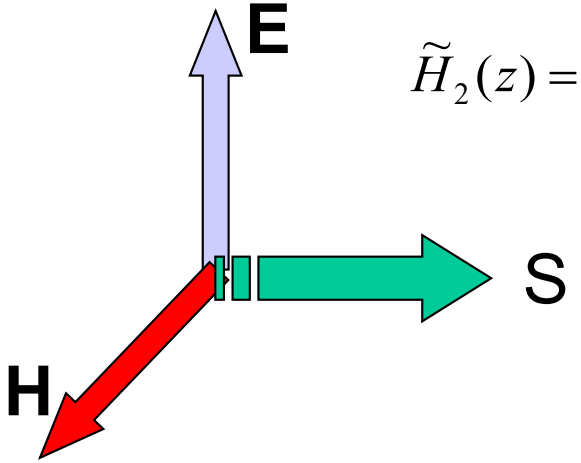
$$\tilde{E}_1(z) = \Gamma\overline{E}_1^+ e^{+\bar{\gamma}_1 z}\hat{x}$$

$$\tilde{H}_1(z) = \frac{-1}{\overline{\eta}_1}\Gamma\overline{E}_1^+ e^{+\bar{\gamma}_1 z}\hat{y}$$



$$\tilde{E}_2(z) = \tau\overline{E}_1^+ e^{-\bar{\gamma}_2 z}\hat{x}$$

$$\tilde{H}_2(z) = \frac{1}{\overline{\eta}_2}\tau\overline{E}_1^+ e^{-\bar{\gamma}_2 z}\hat{y}$$



$$\Gamma = \frac{\overline{E}_1^-}{\overline{E}_1^+} = \frac{\overline{\eta}_2 - \overline{\eta}_1}{\overline{\eta}_1 + \overline{\eta}_2}$$

$$\tau = \frac{\overline{E}_2^+}{\overline{E}_1^+} = 1 + \Gamma = \frac{2\overline{\eta}_2}{\overline{\eta}_1 + \overline{\eta}_2}$$

Next: Transmission lines